

- This exam contains 10 pages (including this cover page) and 16 problems. Check to see if any pages are missing.
- This is the second exam, and counts for 20% of your final grade. You have from 3:35 PM on Thursday (06/20/2019) through 1:35 PM on Monday (06/24/2019) to complete this exam. You may use your notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- If a question is followed by work, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work when work is required will receive very little to no credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. **Work scattered all over the page without a clear ordering will receive very little to no credit.**
- Double check your work to make sure it is free of basic algebra errors. **Solutions which contain a substantial number of basic algebra errors or serious, substantially incorrect algebra errors will receive very little to no credit.**

As this is an open-note take-home exam, the standards used to evaluate your performance will, correspondingly, be higher than those used to evaluate your performance on a regular, in-class exam. **To earn full points on a given problem, you must submit a high-quality, coherent, organized, polished, and substantially correct solution.**

There are 300 points total in this exam, as well as 2 bonus questions worth a total of 40 possible points.

Page:	3	4	5	6	7	8	Total
Points:	80	40	60	50	20	50	300
Score:							

Bonus:	
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1. 20 points Find $C'(x)$ given that the total cost (in hundreds of dollars) of producing x speed-boat hulls is

$$C(x) = 30 + 8(12x^4 + 3)^{-3} + x^{12}.$$

work

2. Let $y = y(x)$ (that is, let y represent a function of x). For the following problems, find $\frac{dy}{dx}$.

(a) 20 points $2x^2 = 40y^2 + 1$

work

(b) 20 points $\frac{6 + 4x^2}{2y^2 - 2} = x^6$

work

(c) 20 points $x^2 \ln(x^6) = e^{3y^3+3}$

work

3. 20 points If $f(x) = e^{(x+7)^{111}}$, find the value(s) of x where the tangent line of f is horizontal.

work

4. Let $x = x(\star)$, $t = t(\star)$ (that is, if both x and t represent functions of $\star$), and suppose that x and t are related by

$$6e^{x^4} = e^2 + (x^2t)^2.$$

- (a) 10 points Find $\frac{dt}{d\star}$

work

- (b) 10 points Find $\frac{dt}{d\star}$ when $x = 2$, $t = 1$, and $\frac{dx}{d\star} = 7$.

work

5. 20 points Find the relative rate of change of $f(x) = 3 + e^x(x - 5)^3$.

work

6. 20 points Given the price-demand equation $x = f(p) = p + (1 - p^3)^4$, find $E(p)$, the price-elasticity of demand. At $p = 0.5$, is demand elastic or inelastic?

work

7. 20 points Given the price-demand equation $x = f(p) = e^{200 - \frac{1}{2}p^2}$, find the value(s) of the price p for which demand has unit elasticity.

work

8. 20 points Given the price-demand equation $x = f(p) = -p^4 + \frac{16}{3}p^3 - \frac{15}{2}p^2$, what price p maximizes the revenue?

(*Hint*: Revenue equals price times demand.)

work

9. 10 points If $f(x) = \frac{x^{-1}}{4^{-1}} - \frac{x^{-2}}{4^{-1}}$, find the intervals on which $f(x)$ is increasing.

work

10. 20 points Let $f(x) = (x-4)^7 e^{-7x}$. Find the intervals on which $f(x)$ is increasing, the intervals on which $f(x)$ is decreasing, and the local extrema (if there are any).

(*Hint*: factor by grouping to find all the roots of your derivative.)

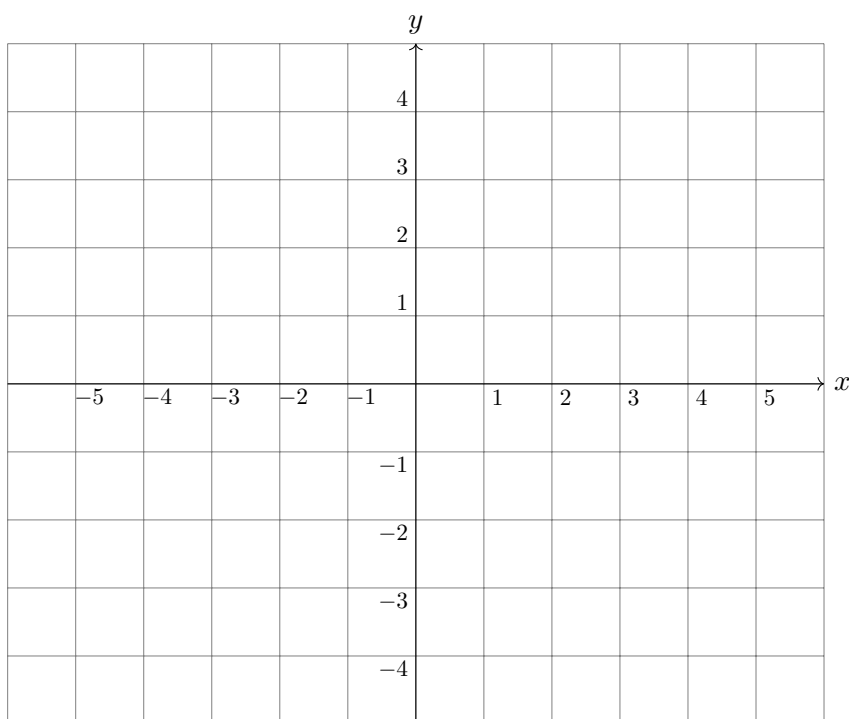
work

11. 20 points Sketch a graph of a single function $f(x)$ which is continuous on $(-\infty, \infty)$ and satisfies:

$$f(2) = 1, \quad f'(-2) = 0, \quad f'(2) = 0, \quad f''(0) = 0, \quad f''(-2) = 0$$

$$f'(x) > 0 \text{ on } (-2, 2) \text{ and } (-\infty, -2), \quad f'(x) < 0 \text{ on } (2, \infty).$$

(*Hint*: think about x^3)



12. 10 points If $y = e^{3x+2} + (2x - 1)^5$, find $\frac{d^2y}{dx^2}$.

work

13. 20 points If $f(x) = e^{7x} - 7e^{\sqrt{7}x} + \frac{7}{6}x^7$, find any values of x for which $f''(x) = 0$.

(Hint: $e^0 = 1$)

work

14. 20 points If $f(x) = 27x^2 - 9x^{-2}$, find the x and y coordinates of all inflection points.

work

15. 20 points (bonus) Let y represent a function of x . Derive a general formula for $\frac{d}{dx} (x^n e^{y^m})$. Show all of your steps.

work

16. 20 points (bonus) Given the price-demand equation $x = f(p) = 50 - p^2$, find the equation of the line tangent to the graph of the elasticity of demand function at $p = 5$.

work